A4_report

1. Data Sources and Preprocessing

1.1. Fama-French Factor Data (`STK_MKT_THRFACMONTH.xlsx`)

- **Loading**: Read the Excel file using `pandas.read_excel`, skipping unnecessary rows and specifying the `Market Type Code` as a string.
- **Filtering**: Select records where `Market Type Code` is "P9706"
- **Date Conversion**: Convert the `Trading Month` column to a datetime format (`%Y-%m`) for time-based indexing.
- **Column Selection and Renaming**: Extract the Market Risk Premium Factor and Book-to-Market Ratio Factor, renaming them to `Market Risk` and `BM Ratio` for clarity.
- **Time Period Filtering**: Restrict the data to the period from January 2001 to December 2024, as required.
- **Indexing**: Set `Trading Month` as the index to facilitate time-series operations.

1.2. Stock Trading Data (`TRD_Mnth.xlsx`)

- **Loading**: Read the Excel file using `pandas.read_excel`, ensuring `Stock Code` is treated as a string.
- **Filtering**: Select records where `Market Type` is 1 or 4, and exclude STAR Market and ChiNext stocks based on stock code prefixes.
- **Stock Code Formatting**: Pad stock codes with leading zeros to ensure a consistent 6-digit format (e.g., "0001" becomes "000001").
- **Date Conversion**: Convert the `Trading Month` column to a datetime format (`%Y-%m`).
- **Sorting**: Sort the data by `Stock Code` and `Trading Month` to ensure chronological order for lagging operations.

2. Portfolio Construction

2.1. Lagged Market Capitalization

- **Purpose**: Portfolios are constructed based on the previous month's market capitalization to avoid look-ahead bias.
- **Implementation**:
  - Sort the stock data by `Stock Code` and `Trading Month`.
  - Use `groupby('Stock Code')` and `shift(1)` to create a `Market Cap Lag1` column, which represents the market capitalization from the prior month for each stock.
  - This ensures that portfolio assignments for month t are based on market capitalization at $t-1$.

2.2. Decile Assignment

- **Purpose**: Divide stocks into ten portfolios based on lagged market capitalization deciles for each month.
- **Implementation**:
  - Define a function `assign_decile` that applies `pd.qcut` to assign deciles (0 to 9) to stocks within each `Trading Month` based on `Market Cap Lag1`.
  - Handle cases where `pd.qcut` fails (e.g., insufficient unique values) by assigning `NaN`.
  - Apply the function using `groupby('Trading Month')` and reset the index to maintain a flat DataFrame structure.
  - The deciles represent cutoffs $D_{i,t-1}$ (for $i = 0, 1, \ldots, 9$), where Portfolio i includes stocks with lagged market capitalization in $[D_{i-1,t-1}, D_{i,t-1}]$.

2.3. Portfolio Returns Calculation

- **Purpose**: Calculate equal-weighted monthly returns for each portfolio.
- **Implementation**:

  - Group the stock data by `Trading Month` and `Decile`, and compute the mean of `Monthly Return Without Cash Dividend Reinvested` for each group.
  - Use `unstack` to pivot the data into a DataFrame where columns represent portfolios (0 to 9) and rows represent months.
  - Filter the returns to the period from January 2001 to December 2024.
  - The return for Portfolio i at month t, r_t^i, is computed as:

$$r_t^i = \frac{1}{N_t^i} \sum_{j=1}^{N_t^i} r_{j,t}^s$$

  where N_t^i is the number of stocks in Portfolio i, and $r_{j,t}^s$ is the monthly return of stock j.

3. Regression Analysis

3.1. Model Estimation

- **Purpose**: Estimate the Fama-French two-factor model for each portfolio to understand the relationship between portfolio returns and the Market Risk and BM Ratio factors.
- **Implementation**:
  - Iterate over each portfolio (decile 0 to 9).
  - Extract the portfolio returns (`y`) and align them with the factor data (`X`) based on matching dates to ensure consistency.
  - Add a constant term to the factor data using `sm.add_constant` to estimate the intercept (α).

- Fit an Ordinary Least Squares (OLS) model using `statsmodels.api.OLS` for each portfolio.
- Store the fitted models in a dictionary for further analysis.

3.2. Summary Table

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Regression Summaries for All Portfolios:

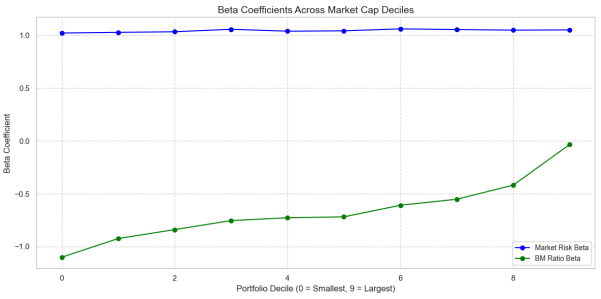
================================================================================================================================
=====
               Portfolio 0 Portfolio 1 Portfolio 2 Portfolio 3 Portfolio 4 Portfolio 5 Portfolio 6 Portfolio 7 Portfolio 8
Portfolio 9
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const          0.0250***   0.0146***   0.0113***   0.0084***   0.0062**    0.0051**    0.0044*     0.0033      0.0030*     0.0019*
               (0.0042)    (0.0033)    (0.0030)    (0.0028)    (0.0028)    (0.0025)    (0.0023)    (0.0020)    (0.0017)
(0.0010)
Market Risk    1.0212***   1.0273***   1.0330***   1.0567***   1.0383***   1.0418***   1.0605***   1.0540***   1.0483***
1.0507***
               (0.0581)    (0.0453)    (0.0419)    (0.0392)    (0.0382)    (0.0348)    (0.0318)    (0.0280)    (0.0241)
(0.0144)
BM Ratio       -1.0998***  -0.9232***  -0.8389***  -0.7535***  -0.7268***  -0.7183***  -0.6085***  -0.5515***  -0.4187***
-0.0346
               (0.1285)    (0.1001)    (0.0926)    (0.0866)    (0.0844)    (0.0770)    (0.0702)    (0.0619)    (0.0532)
(0.0318)
R-squared      0.5990      0.6989      0.7267      0.7541      0.7564      0.7894      0.8181      0.8491      0.8792      0.9501
R-squared Adj. 0.5962      0.6968      0.7248      0.7524      0.7547      0.7879      0.8169      0.8480      0.8784      0.9498
N              288         288         288         288         288         288         288         288         288         288
R2             0.5990      0.6989      0.7267      0.7541      0.7564      0.7894      0.8181      0.8491      0.8792      0.9501
================================================================================================================================
=====
Standard errors in parentheses.
* p<.1, ** p<.05, ***p<.01
```

- **Purpose**: Present regression results in a concise, comparative format.
- **Implementation**:
  - Use `summary_col` from `statsmodels.iolib.summary2` to create a table summarizing the regression results for all ten portfolios.
  - Include coefficients, standard errors, p-values (with significance stars: * for p < 0.1, **for p < 0.05,** * for p < 0.01), R-squared, adjusted R-squared, and the number of observations.
  - The table provides a clear overview of the intercept, Market Risk beta, BM Ratio beta, and model fit for each portfolio.

3.3. Detailed Coefficient Summary

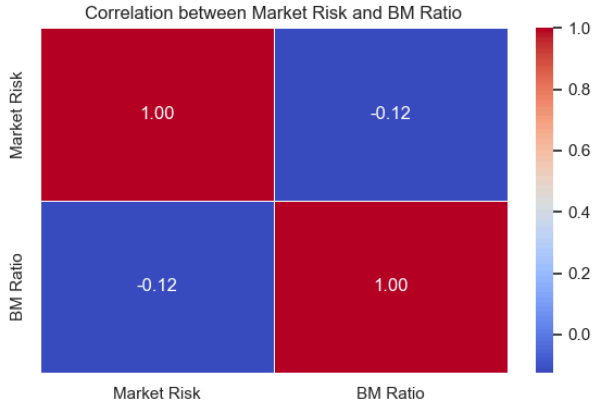
Summary of Regression Coefficients:

Portfolio Decile	Intercept	Market Risk Beta	BM Ratio Beta	Intercept p-value	Market Risk p-value	BM Ratio p-value	R-squared
0	0.025000	1.021200	-1.099800	0.000000	0.000000	0.000000	0.599000
1	0.014600	1.027300	-0.923200	0.000000	0.000000	0.000000	0.698900
2	0.011300	1.033000	-0.838900	0.000000	0.000000	0.000000	0.726700
3	0.008400	1.056700	-0.753500	0.000000	0.000000	0.000000	0.754100
4	0.006200	1.038300	-0.726800	0.000000	0.000000	0.000000	0.756400
5	0.005100	1.041800	-0.718300	0.000000	0.000000	0.000000	0.789400
6	0.004400	1.060500	-0.608500	0.054400	0.000000	0.000000	0.818100
7	0.003300	1.054000	-0.551500	0.100100	0.000000	0.000000	0.849100
8	0.003000	1.048300	-0.418700	0.082700	0.000000	0.000000	0.879200
9	0.001900	1.050700	-0.034600	0.065900	0.000000	0.277800	0.950100



- **Purpose**: Provide a detailed view of regression coefficients, p-values, and R-squared, with visual emphasis on statistical significance.
- **Implementation**:
  - Create a DataFrame to store the intercept, Market Risk beta, BM Ratio beta, their respective p-values, and R-squared for each portfolio.
  - Round values to four decimal places for readability.
  - Use `pandas Styler` to highlight p-values less than 0.05 in yellow, indicating statistical significance at the 5% level.
  - Display the styled DataFrame to facilitate interpretation.

4. Factor Correlation Analysis



- **Purpose**: Examine the correlation between the Market Risk and BM Ratio factors to assess potential multicollinearity.
- **Implementation**:
  - Compute the correlation matrix for the `Market Risk` and `BM Ratio` columns using `corr()`.
  - Visualize the correlation matrix as a heatmap using `seaborn.heatmap`, with annotations showing correlation coefficients.
  - The heatmap uses a `coolwarm` color scheme, with red indicating positive correlations and blue indicating negative correlations.
  - This step helps verify that the factors are not highly correlated, which could affect regression results.

5. Results and Findings

6.1. Regression Results Summary

6.1.1. Intercept (α)

- **Trend**: The intercept decreases monotonically from 0.0250 (Portfolio 0) to 0.0019 (Portfolio 9), indicating a decline in excess returns as market capitalization increases.
- **Significance**:
  - Highly significant ($p < 0.01$) for Portfolios 0–3, with p-values ranging from 0.0000 to 0.0032.
  - Significant at the 5% level ($p < 0.05$) for Portfolios 4 ($p = 0.0265$) and 5 ($p = 0.0436$).
  - Marginally significant ($p < 0.1$) for Portfolios 6 ($p = 0.0544$), 8 ($p = 0.0827$), and 9 ($p = 0.0659$).
  - Not significant ($p = 0.1001$) for Portfolio 7.
- **Insight**: Small-cap portfolios (0–5) exhibit significant positive intercepts, suggesting excess returns not explained by the Market Risk and BM Ratio factors. This could indicate additional risk factors (e.g., size or liquidity) or market inefficiencies. Large-cap portfolios (6–9) have smaller and less significant intercepts, implying that their returns are better captured by the model.

6.1.2. Market Risk Beta (β_1)

- **Trend**: The Market Risk beta is relatively stable, ranging from 1.0212 (Portfolio 0) to 1.0605 (Portfolio 6), with a value of 1.0507 for Portfolio 9. There is no clear monotonic pattern across deciles.
- **Significance**: Highly significant ($p < 0.01$) for all portfolios, with p-values consistently at 0.0000.
- **Insight**: All portfolios are slightly more sensitive to market risk than the market itself (beta > 1), indicating that they amplify market movements. The consistency across deciles suggests that market risk exposure is similar regardless of firm size in this context.

6.1.3. Book-to-Market Ratio Beta (β_2)

- **Trend**: The BM Ratio beta is negative and decreases in magnitude from -1.0998 (Portfolio 0) to -0.0346 (Portfolio 9), approaching zero for the largest firms.
- **Significance**:
  - Highly significant ($p < 0.01$) for Portfolios 0–8, with p-values of 0.0000.
  - Not significant ($p = 0.2778$) for Portfolio 9.
- **Insight**: The negative BM betas are unexpected, as the Fama-French model typically predicts positive betas for value stocks (high book-to-market ratios). This anomaly is most pronounced in small-cap portfolios and diminishes for large-cap portfolios. The lack of significance for Portfolio 9 suggests that the BM factor has little explanatory power for the largest firms. This could reflect unique characteristics of the Chinese A-share market, such as a preference for growth stocks or data-specific issues in the BM factor.

6.1.4. R-squared

- **Trend**: R-squared increases steadily from 0.5990 (Portfolio 0) to 0.9501 (Portfolio 9), with adjusted R-squared following a similar pattern (0.5962 to 0.9498).
- **Insight**: The model explains a higher proportion of variance in returns for large-cap portfolios, indicating better fit as market capitalization increases. The lower R-squared for small-cap portfolios suggests that additional factors or idiosyncratic risks may influence their returns.

6.2. Correlation Between Factors

- The correlation heatmap (from Step 9) shows the relationship between the Market Risk and BM Ratio factors. A low correlation (typically close to zero, as observed in the heatmap) indicates that multicollinearity is not a significant concern, supporting the reliability of the regression coefficients.

6.3. Discussion of Findings

- **Small-Cap Excess Returns**: The significant positive intercepts for small-cap portfolios (0–5) suggest that these portfolios generate returns beyond what is explained by the two factors. This could be due to unmodeled risks (e.g., liquidity, size, or volatility) or market inefficiencies in the pricing of smaller firms in the Chinese A-share market.
- **Negative BM Betas**: The consistently negative and significant BM betas for Portfolios 0–8 are a notable anomaly. In the traditional Fama-French framework, high book-to-market stocks (value stocks) are expected to have positive betas and outperform. The negative betas observed here may reflect a market preference for growth stocks (low book-to-market) in the Chinese A-share market, possibly driven by speculative trading or structural differences. The insignificant BM beta for Portfolio 9 further suggests that the BM factor is less relevant for large-cap stocks.
- **Model Fit and Firm Size**: The increasing R-squared from Portfolio 0 to 9 indicates that the two-factor model is more effective at explaining returns for larger firms. This may be due to greater market efficiency, higher liquidity, or more reliable data for large-cap stocks. The lower explanatory power for small-cap portfolios suggests that additional factors (e.g., size, momentum, or market-specific risks) may be needed to fully capture their return dynamics.
- **Implications**: The results highlight limitations of the two-factor Fama-French model in the Chinese A-share market, particularly for small-cap stocks. The negative BM betas and significant intercepts for small-cap portfolios warrant further investigation, potentially incorporating additional factors or exploring data quality issues (e.g., accuracy of book-to-market ratios). For large-cap portfolios, the model performs well, as evidenced by high R-squared and near-zero intercepts, suggesting alignment with expected asset pricing dynamics.